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AI-Powered Business Analytics System for Performance Evaluation, Insight Extraction, and Strategy Optimization

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Declaration Form

This initial report presents the foundation of the thesis project entitled *AI-Powered Business Analytics System for Performance Evaluation, Insight Extraction, and Strategy Optimization*. We declare that the background study, problem formulation and proposed methodology presented here are the work of the authors under the supervision of Dr. Saifuddin Md. Tareeq. Ideas, methods, datasets and findings originating from prior work are cited. The implementation, experimental results and conclusions are intentionally reserved for later stages of the thesis. This report is current to 7 September 2026; the project remains in progress until February 2027.

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Abstract

Acknowledgements

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Chapter 1

Introduction

1.1 Context and Motivation

Business intelligence and analytics organize operational data so that decision makers can describe performance, anticipate future conditions and choose an appropriate response [6]. For an inventory-based small or medium enterprise (SME), this progression begins with familiar records: invoice lines, product movements, purchase receipts, expenses, payments and customer visits. These records can explain revenue and stock position, but their value is limited when they remain divided among paper ledgers, spreadsheets and separate applications. The owner may know that sales declined last month without knowing which products, customers or operating conditions caused the decline, whether the pattern is likely to continue, or which affordable action should be taken.

The decision is more demanding than producing a dashboard. Before Ramadan or Eid, for example, a retailer must estimate demand early enough to order from a supplier. Ordering too little creates stock-outs and forgone sales; ordering too much immobilizes working capital and may create obsolete stock. At the same time, a previously valuable customer may be becoming inactive, a product may appear profitable only because costs were recorded incompletely, or a promotion may increase volume without increasing contribution. These questions draw on the same operational history but require different analytical tasks and a common decision context.

Retail forecasting research identifies hierarchy, intermittency, promotions, new products, changing assortment and the organizational use of forecasts as central practical concerns [11]. Evidence from large forecast competitions also cautions against assuming that a more complex algorithm is automatically more accurate: combinations, statistical methods and carefully constructed baselines remain competitive across many series [25, 26]. The relevant research problem is therefore not to attach a fashionable model to a sales file. It is to construct a defensible path from business events to out-of-time predictions and then from predictions to feasible, explainable actions.

1.2 Bangladesh SME Context

The intended setting is an inventory-holding retail SME in Bangladesh. This scope is narrower than the approved thesis title, but it makes the business entities, prediction units and decisions testable. Bangladesh-specific studies show that ICT adoption in SMEs is shaped by perceived benefit, management and government support, cost, complexity and the surrounding organizational and technological conditions [17, 32]. Research on mobile financial services further shows that users value accessibility, mobility, self-control and other affordances embedded in everyday practice [14]. These findings imply that local payment channels, low-friction data entry and owner-readable explanations are design requirements, not optional presentation details.

This study uses a functional definition of its population rather than relying on a regulatory size threshold that may change. A target business holds stock, records sales and purchases, makes branch- or product-level decisions, and has limited specialist analytics capacity. The unit of analysis may be a business, branch, stock-keeping unit (SKU), customer or historical customer snapshot, depending on the research question. Conclusions will be restricted to the participating firms and the evidence actually collected.

1.3 Problem Statement

The central problem is the absence of a trustworthy and usable connection among three activities: recording SME operations, extracting predictive insight and selecting a feasible business action. A transaction system can preserve orders and stock but still provide no forecast. A standalone model can report an accuracy score but still be unusable because its features would not exist at the decision time, its test data leaked information from training, or its output is not linked to budget, stock, lead time and customer consent. A recommendation can be technically valid yet operationally impossible.

The problem is decomposed into four linked gaps:

- G1. **Data foundation gap.** Sales, inventory, purchases, expenses, customers and payments need a consistent schema, timestamps, provenance and organization boundaries before they can support comparable analytics.
- G2. **Prediction-validity gap.** Demand and customer-risk models must use historical snapshots, chronological or entity-safe splits, training-only transformations, strong baselines and metrics appropriate to sparse or imbalanced outcomes. Leakage can otherwise create persuasive but invalid performance [21].
- G3. **Decision gap.** A forecast or risk probability does not specify what to order, whom to contact or whether the action is affordable. Prediction must be connected to costs,

constraints and uncertainty; this is the transition from predictive to prescriptive analytics [1].

- G4. **Adoption and accountability gap.** An SME owner needs a concise reason, confidence indication and traceable source for a suggestion. Feature attribution can describe model behavior, but it neither proves causality nor replaces human judgment [24, 27].

Accordingly, this thesis asks how an integrated, Bangladesh-aware framework can transform auditable retail records into leakage-controlled forecasts and customer insights, and then rank actions that respect the operating constraints of an SME. The work does not presume that machine learning will defeat every baseline. Discovering that a simple rule is more reliable or economical for a particular task is a valid outcome.

1.4 Research Aim and Objectives

1.4.1 General Objective

The general objective is to design and evaluate an AI-powered business analytics framework that enables Bangladeshi retail SMEs to monitor performance, extract reliable and explainable insights, and prioritize feasible strategies.

1.4.2 Specific Objectives

The study will:

- O1. define an auditable canonical data model for sales, products, stock, purchases, returns, customers, suppliers, payments and expenses;
- O2. construct reproducible, time-indexed preprocessing and feature pipelines for performance analysis, SKU demand, customer inactivity and segmentation;
- O3. compare simple, statistical and machine-learning approaches under chronological or entity-safe validation, using both predictive and decision-oriented measures;
- O4. quantify uncertainty and provide global and case-level supporting factors without presenting association as causation;
- O5. formulate a constraint-aware mechanism for reorder, retention and operational action ranking; and
- O6. evaluate technical accuracy, robustness, scalability, usability and potential business value on clearly separated evidence tiers, culminating in consenting Bangladeshi SME data.

1.5 Initial Solution Concept

The proposed solution is B-SMART: the Bangladesh SME Monitoring, Analytics, Recommendation and Tracking framework. It is an end-to-end decision-support process rather than a new base classifier. Business events enter a governed operational layer; validated events are transformed into time-frozen analytical views; candidate models generate forecasts, risks and segment profiles; and a strategy layer removes infeasible actions before ranking the remainder. Each displayed action carries its data cutoff, model or baseline source, uncertainty and supporting factors. Owner response and later outcome are retained for evaluation and monitoring.

Three analytical outputs match the approved title. *Performance evaluation* summarizes revenue, margin, stock movement and customer behavior from reconciled records. *Insight extraction* estimates future demand and customer inactivity and identifies meaningful behavioral groups. *Strategy optimization* converts those estimates into candidates such as a reorder quantity or retention priority, subject to cash, storage, lead time, minimum order quantity, current stock and contact consent. Chapter 3 defines these components formally.

1.6 Research Questions

The thesis is guided by four questions:

RQ1.

Do calendar, price, promotion and market-context features improve out-of-time branch-SKU demand forecasts over seasonal and statistical baselines in the target SME setting?

RQ2.

How accurately and reliably can a historical customer snapshot identify no repeat purchase during a fixed future horizon?

RQ3.

Under the same demand and operating constraints, does a forecast-driven reorder policy reduce stock-out and holding cost relative to the participating business's current policy?

RQ4.

Do concise Bangla action explanations improve owner task completion, confidence and perceived usability without creating excessive decision time?

1.7 Scope and Boundaries

The research covers retail and inventory-based SMEs, structured operational data, descriptive indicators, demand forecasting, customer inactivity, behavioral segmentation, explanation and two strategy families: replenishment and consent-aware customer retention. The initial geographical context is Bangladesh; an external dataset may test method portability but cannot establish Bangladeshi business behavior. Public price data may supply market context but cannot substitute for invoice-level evidence.

The study excludes manufacturing scheduling, credit scoring, employee surveillance, fully autonomous price changes and unrestricted customer profiling. It does not claim causal effects from observational feature attributions. It will not treat generated data as field evidence, combine synthetic and real observations in a reported real-world score, or generalize from one retailer to all SMEs. Implementation detail, experimental results and final conclusions are deliberately reserved for Chapters 4–6.

1.8 Expected Contributions and Significance

The expected research contribution is the evaluated integration of established methods under constraints that matter to small retailers. Specifically, the work is expected to provide: (i) a traceable operational-to-analytical data design; (ii) reusable historical-snapshot protocols that prevent common forms of leakage; (iii) a Bangladesh-aware comparison of forecasting and customer analytics baselines; (iv) a transparent bridge from prediction to constrained action; and (v) evidence about whether owners can correctly interpret and use the resulting recommendations.

The practical significance is equally bounded. The proposed system is intended to help an owner see the evidence behind a decision, compare a recommendation with a familiar rule and retain authority over the final action. It is not intended to remove the owner from the decision or to label a statistically accurate output as profitable before business impact is measured.

1.9 Anticipated Challenges

The most consequential dependency is suitable primary data. The intended joint history of invoice lines, stock, products, prices and customers is difficult to obtain from a public Bangladesh source, and partner data may contain missing identifiers, changing SKU codes or incomplete cost records. Technical risks include intermittent demand, short histories, new products, stock-out censoring, class imbalance and drift. Scientific risks include target leakage, optimistic model selection, incompatible dataset comparisons and conclusions based on one firm. Human risks include time pressure, varied digital skill, low trust

and over-trust in confident-looking explanations. The methodology assigns a specific prevention or evaluation step to each of these risks.

1.10 Organization of the Report

Chapter 2 develops the technical and contextual background, reviews the verified literature and identifies the research gap. Chapter 3 presents the research design, data protocol, formal tasks, proposed framework, evaluation plan and two-semester timeline. The Implementation, Experimental Results and Conclusions chapters remain as named placeholders in this initial report and will be completed only when the corresponding work and evidence are available.

Chapter 2

Background Study and Related Work

2.1 Background Study

2.1.1 Business Intelligence and the Decision Continuum

Business intelligence (BI) combines data, infrastructure, analytical methods and organizational practice to improve decisions. Chen, Chiang and Storey trace its development from structured enterprise reporting toward web, mobile and sensor-rich analytics [6]. This thesis adopts a three-stage decision continuum. *Descriptive analytics* establishes what happened and where; *predictive analytics* estimates what may happen next; and *prescriptive analytics* compares feasible responses. The stages are dependent: an action recommendation is only as credible as the prediction that supports it, and a prediction is only as credible as the operational data from which it was derived.

Operational and analytical workloads also have different integrity needs. A sale requires a consistent stock and payment update, whereas a model requires a historical feature view whose values do not change after evaluation. The proposed framework therefore separates the transaction record from versioned analytical snapshots. It retains the relationship between them through source identifiers, timestamps and transformation metadata.

2.1.2 SME Digitalization in Bangladesh

Technology adoption is not determined by model accuracy alone. A mixed-method case study of rural Bangladeshi SMEs identifies awareness of benefits, government support, top-management support and financial support as important ICT-adoption determinants [17]. A later survey of 535 SME managers combines the technology–organization–environment

framework with diffusion-of-innovation constructs; relative advantage, complexity, perceived cost, management factors, competitive pressure and government support are among the significant determinants reported [32]. These studies justify attention to cost, workflow fit and organizational readiness, although neither evaluates a retail prediction system.

Mobile financial services are part of this operating context. Interview-based research in Bangladesh identifies accessibility, spatial and temporal mobility, self-control and related affordances in everyday MFS use [14]. Consequently, the proposed data model treats payment mode and settlement status as ordinary business fields. The research does not infer that a particular provider causes better performance; it asks whether locally recognizable records and explanations make the analytical workflow more usable.

2.1.3 Performance Evaluation and Data Foundations

Performance evaluation begins with well-defined measures. Revenue, gross margin, average order value, stock turnover, return rate and customer-repeat rate must each specify a time window, aggregation level and treatment of cancellations, refunds and missing costs. A dashboard value without these definitions can disagree with the accounting record or compare unlike periods. The canonical layer therefore preserves invoice lines and stock movements while deriving analytical measures in documented views.

Data provenance answers where a record came from, when it was received and how it was transformed. For primary SME data, provenance must be joined with organization isolation and pseudonymous customer identifiers. For public data, the license, original context and permissible inference must be recorded. A foreign transaction dataset can test whether code handles real invoices but cannot establish Bangladesh-specific customer behavior.

2.1.4 Customer Analytics: RFM, CLV and Segmentation

RFM summarizes *recency*, *frequency* and *monetary value* from a customer's transaction history. These features are inexpensive to derive and retain a clear behavioral interpretation. Fader, Hardie and Lee connect RFM to customer lifetime value (CLV) through Pareto/NBD and gamma-gamma components and show how different histories can have similar expected value [10]. CLV is useful as an outcome or planning quantity, but a field used to define a segment cannot also be admitted as an independent predictor of that same segment.

Unsupervised segmentation asks whether coherent behavioral groups emerge without an existing label. For K-Means, the objective is

$$\underset{\{C_k, \mu_k\}_{k=1}^K}{\text{minimize}} \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|_2^2. \quad (2.1)$$

Because the procedure returns clusters even when the data form a continuum, cluster number and interpretation require validation. The silhouette coefficient compares within-cluster cohesion with separation from the nearest other cluster [30]. Stability under re-sampling and business usefulness are separate tests; an attractive label such as “loyal” does not establish that a cluster is statistically stable or actionable.

2.1.5 Retail Demand Forecasting

A demand forecast estimates the units required for a defined product, location and future horizon. Retail series may be affected by price, promotion, stock-outs, festivals, assortment changes and aggregation across products or branches. Fildes, Ma and Kolassa emphasize that retail forecasting practice cannot be reduced to a single algorithm because data hierarchy, operational process and forecast use matter alongside statistical accuracy [11].

Intermittent SKU demand contains many zero periods separated by nonzero demand. Croston’s method separately smooths nonzero size and the interval between demands [9]; Syntetos and Boylan analyze bias and accuracy in intermittent-demand estimates [34]. These works motivate an intermittent-demand baseline instead of forcing every SKU through a dense-series model. At the same time, forecasts must reconcile across product, category, branch and business totals. Optimal-combination reconciliation offers a principled way to produce coherent hierarchical forecasts [18].

Model complexity should follow the evidence. ARIMA-family models represent serial dependence after transformation; tree ensembles can learn nonlinear effects from lagged, calendar and price variables; LSTM networks use gated memory for sequential dependence [16]. Random Forest and XGBoost provide strong tabular comparators [2, 7]. However, forecast competitions show that statistical, combined and hybrid approaches remain highly competitive across large collections [25, 26]. A seasonal-naïve forecast is therefore a required benchmark, not a ceremonial baseline.

Forecast accuracy and inventory usefulness are related but not identical. Hyndman and Koehler demonstrate defects in common percentage errors and propose MASE for scale-free comparison [19]. Syntetos and colleagues connect forecast selection with stock-control performance in a wholesaling study [33]. The methodology accordingly reports point error, uncertainty and simulated stock-out/holding cost rather than selecting a method from one convenient metric.

2.1.6 Customer Inactivity, Class Imbalance and Calibration

In a subscription service, churn may be an observed cancellation. In non-contractual retail it must be defined at a cutoff t : no repeat purchase during a fixed interval $(t, t + H]$. Every feature must be known by t , and the future label window must not contaminate training

or validation. This definition turns an ambiguous business word into a reproducible prediction task.

The positive class may be rare or change across businesses. Burez and Van den Poel show that sampling and cost-sensitive treatments affect churn modelling under class imbalance [3]. Saito and Rehmsmeier explain why precision–recall analysis is more revealing than ROC analysis for imbalanced binary tasks [31]. Verbeke and colleagues further show that the best discriminating churn model need not be the most profitable when contact cost and benefit are considered [35].

Ranking is not enough when the output is used in a decision calculation. A probability of 0.7 should correspond, approximately, to the event frequency among comparable cases. Niculescu-Mizil and Caruana compare probability estimates and post-hoc calibration methods for supervised learners [29]. The proposed framework therefore reports Brier score and calibration in addition to PR-AUC, ROC-AUC and top-fraction lift.

2.1.7 Leakage, Explainability and Change Over Time

Data leakage arises when training has access to information that would not legitimately exist at prediction time. It may enter through a target-derived field, a post-outcome variable, repeated customers in both sides of a split or a transform fitted before the split [21]. Leakage control is therefore a property of the entire data pipeline, not a model option. A clear prediction time, feature availability table and split policy precede training.

SHAP represents a prediction as additive feature contributions relative to a reference value [24]; efficient tree-specific methods support local and global inspection [23]. Such an attribution describes how a fitted model used encoded features. Miller’s synthesis of social science research emphasizes that useful explanations are selective, contextual and often contrastive [27]. Thus a recommendation card should answer “why this action now?” in a small number of decision-relevant factors, while preserving access to fuller evidence and avoiding causal wording.

Business conditions also change. Concept drift occurs when the relationship between predictors and outcomes changes over time; Gama and colleagues organize adaptation methods and evaluation strategies for this setting [12]. Retail monitoring must distinguish data-quality failure, input-distribution shift and performance decay. A retraining schedule alone is not evidence that a model remains valid.

2.1.8 From Prediction to Constrained Action

Prescriptive analytics uses predictions within an explicit decision problem. Bertsimas and Kallus formulate data-driven prescriptions in which auxiliary observations inform conditional stochastic optimization [1]. The present thesis adopts the underlying principle—evaluate actions by expected benefit, cost and constraints—without claiming to reproduce

that paper’s asymptotic method. Reorder and retention are kept as separate action families because their outcomes, costs, permissions and uncertainties differ.

The complete framework is an information-system artifact as well as an analytical study. Design-science research builds and evaluates artifacts in relation to a relevant problem and a knowledge base [15]. This supports an evaluation plan that includes predictive validity, software behavior, owner usability and decision consequences instead of reporting model accuracy as the sole measure of success.

2.2 Literature-Review Procedure

The present review is a structured background study, not a claim of an exhaustive systematic review. Sources were selected for direct relevance to one of six themes: Bangladesh SME digitalization, BI and design science, customer analytics, retail demand/inventory, churn and evaluation integrity, and explainable or prescriptive analytics. Searches used combinations of terms such as *Bangladesh SME ICT adoption*, *retail demand forecasting*, *intermittent demand*, *RFM customer segmentation*, *churn profit*, *data leakage*, *probability calibration*, *concept drift* and *prescriptive analytics*.

Peer-reviewed journal or conference work was preferred. A source was retained when it defined a required concept, supplied a method or evaluation principle, or provided Bangladesh-specific evidence. Metadata for every DOI-bearing entry was checked against Crossref, and dataset records were checked against DataCite or the official repository. Specific numerical claims were included only when supported by the paper metadata, abstract or full record available to the authors. The core set contains 33 scholarly works; candidate datasets are listed separately in Chapter 3.

2.3 Thematic Synthesis of Prior Work

2.3.1 Business and Bangladesh Evidence

The BI literature supplies the descriptive–predictive–prescriptive frame [6], while design science supplies an artifact-oriented research logic [15]. Bangladesh studies explain why organizational support, perceived value, cost, complexity and actual local use should influence the proposed design [14, 17, 32]. Their limitation for the present problem is not quality but unit of analysis: adoption surveys and interviews do not provide invoice-level evidence for demand, inactivity or inventory decisions.

2.3.2 Customer Evidence

Ngai, Xiu and Chau classify 87 CRM data-mining studies selected from a much larger screened body and show the breadth of classification, association and retention applications [28]. Fader et al. provide a probabilistic link between transaction history and future value [10]; Kim et al. connect customer value with segment-specific strategy [22]; and Cheng and Chen combine RFM, clustering and rough-set classification for customer-value segmentation [8]. Together they motivate behavior-based profiles, but they do not by themselves establish a leakage-safe, non-contractual inactivity model for Bangladeshi retail.

2.3.3 Forecasting and Inventory Evidence

Carbonneau, Laframboise and Vahidov compare machine-learning and traditional approaches using simulated supply-chain data and real foundry orders; their results reinforce the need to compare advanced methods with regression and other simple alternatives [4]. M4 and M5 broaden the evidence across large and retail-specific collections [25, 26]. Retail review evidence [11], intermittent-demand methods [9, 34], hierarchical reconciliation [18] and stock-control evaluation [33] show why an SME comparison requires more than an LSTM and one error column.

2.3.4 Risk, Integrity and Action Evidence

Churn studies establish imbalance-aware and profit-aware evaluation [3, 35]; precision-recall analysis and probability calibration address two different properties of a risk output [29, 31]. Leakage research explains why apparently excellent results can be unusable [21]. SHAP, human-centred explanation and drift research then address inspection, communication and change [12, 23, 24, 27]. Finally, prescriptive analytics motivates placing predictions inside a cost-and-constraint model [1]. The unresolved question is whether these principles can be joined into a low-resource SME workflow and evaluated on the same temporal evidence chain.

2.4 Literature Matrix

Table 2.1 condenses the core set. “Relevant boundary” identifies what remains outside a source’s own objective; it is not a criticism of that work.

TABLE 2.1: Literature matrix for the 33-work core scholarly set.

Ref.	Context or data	Method	Main contribution used here	Relevant boundary for this thesis
[6]	Cross-domain BI research	Framework and bibliometric review	Descriptive-to-predictive BI evolution	No local SME artifact evaluation
[15]	Information-systems research	Design-science framework	Build-and-evaluate research logic	Does not define the retail analytics solution
[17]	Rural Bangladesh SMEs	Case study and questionnaire analysis	Local ICT-adoption determinants	No transaction-level prediction task
[32]	535 Bangladesh SME managers	TOE/DOI model with PLS-SEM	Organizational and technology determinants	Adoption intent, not business impact
[14]	Bangladesh MFS users	Interviews and thematic analysis	Everyday MFS affordances	User context rather than retail forecasting
[28]	87 selected CRM studies	Literature taxonomy	Customer-lifecycle/data-mining map	Predates current temporal and XAI practice
[10]	CDNOW customer cohort	Pareto/NBD and gamma-gamma	Formal RFM-CLV relationship	No stock or action constraints
[22]	Wireless-service customers	CLV-based segmentation	Segment-specific strategy design	Contractual sector and different context
[8]	Customer-value data	RFM, K-Means and rough sets	Interpretable value classes	No out-of-time action evaluation
[30]	General clustering problems	Silhouette coefficient	Internal cluster validation	Does not establish business usefulness
[2]	General prediction benchmarks	Random Forest	Robust tree ensemble	Probabilities/actions require further checks
[7]	Large sparse benchmark tasks	Regularized tree boosting	Scalable nonlinear tabular learning	Does not prevent leakage by itself
[16]	Long-lag sequence tasks	Gated recurrent network	Learning long dependencies	Artificial tasks do not guarantee SME accuracy
[4]	Simulated supply chain and foundry orders	ML versus traditional forecasting	Direct baseline comparison	Different sector and decision conditions
[19]	Forecast evaluation examples	Accuracy-measure analysis	MASE and metric cautions	No inventory or owner outcome
[25]	100,000 series	Forecast competition	Large-scale comparative evidence	Heterogeneous, not SME-specific
[26]	Hierarchical Walmart retail series	Retail forecast competition design	Hierarchy, scale and uncertainty setting	Resource level unlike small shops
[11]	Retail research and practice	Critical review	Retail-specific data/process agenda	Does not implement the proposed local system
[9]	Intermittent stock demand	Separate size/interval smoothing	Foundational intermittent-demand method	Requires modern comparative evaluation
[34]	Intermittent-demand estimates	Bias and accuracy analysis	Corrected evaluation of estimators	No Bangladesh retail test
[18]	Hierarchical time series	Forecast reconciliation	Coherent forecasts across aggregation levels	Does not select business actions
[33]	Wholesaling demand and inventory	Forecasting/stock-control study	Links forecast choice to stock performance	Different organization and cost structure
[13]	Probabilistic prediction	Proper scoring-rule theory	Principled distributional evaluation	Requires a domain decision layer

Ref.	Context or data	Method	Main contribution used here	Relevant boundary for this thesis
[3]	Service churn	Sampling and cost-sensitive models	Imbalance-aware churn comparison	Contractual churn differs from retail inactivity
[35]	Telecom churn datasets	Profit-driven model selection	Discrimination need not maximize profit	SME benefit/cost must be estimated
[31]	Imbalanced binary problems	ROC/PR analysis	Value of precision–recall evaluation	Not a retail modelling study
[29]	Supervised benchmark datasets	Probability calibration comparison	Reliable probabilities as separate objective	Temporal drift remains outside evaluation
[21]	Competition and industry cases	Leakage formulation and taxonomy	Detection and prevention principles	Domain-specific availability must be mapped
[24]	Predictive models	Shapley additive explanations	Unified local feature attribution	Attribution is not causal evidence
[23]	Tree-model applications	Efficient tree explanations	Local-to-global inspection	Inherits model and data defects
[27]	Social science and AI literature	Interdisciplinary review	Selective, contrastive explanation principles	Requires owner-centred validation
[12]	Streaming supervised learning	Concept-drift survey	Adaptation and evaluation taxonomy	Monitoring must be tailored to SME volume
[1]	Data-driven decision problems	Conditional stochastic optimization	Predictive-to-prescriptive bridge	Proposed thesis uses a simpler constrained layer

2.5 Limitations of Existing Approaches

TABLE 2.2: Recurring limitations and the proposed methodological response.

Recurring limitation	Consequence	Response proposed in B-SMART
Static transaction re-reporting	Describes a problem after it occurs	Time-indexed forecasts and risk estimates tied to operational records
Model-only study	Score is disconnected from stock, cash and workflow	Constrained action generation and outcome tracking
Random temporal split	Future or repeated-entity information inflates results	Historical cutoffs, grouped splits and purged outcome windows
Accuracy-only selection	Good average error may produce poor inventory/customer decisions	Calibration, lift, uncertainty and business-cost evaluation
One model for every series	Intermittent, short and dense demand are treated alike	Baseline ladder and data-volume/series-type eligibility rules
Incoherent aggregation	SKU forecasts disagree with category or branch totals	Hierarchical reconciliation when the data support it
Black-box output	Owner cannot judge why action is suggested	Selective supporting factors, source and confidence statement
Foreign or generated evidence	Local validity is overstated	Explicit evidence tiers and claim restrictions
Unmonitored deployment	Performance decay and changed inputs remain hidden	Data-quality, drift, calibration and outcome monitoring

2.6 Research Gap and Thesis Positioning

The gap is derived by comparing what each literature stream establishes with what the stated SME decision problem requires. It is deliberately bounded to the 33-work core set; the thesis does not claim that no related system exists anywhere. Six connected gaps emerge.

G1—Contextual and empirical gap. Bangladesh research explains determinants and lived affordances of digital adoption [14, 17, 32], but it does not provide a joint, longitudinal test of SKU demand, customer inactivity and constrained retail actions. Conversely, large retail and forecast studies offer strong analytical evidence [11, 26], but their scale, market and infrastructure do not establish validity for owner-managed Bangladeshi SMEs. Local adoption evidence and transaction-level analytical evidence therefore remain disconnected.

G2—Operational-to-analytical integration gap. BI and CRM research show how operational data can support multiple analytical functions [6, 28]. The forecasting, segmentation and churn studies in the matrix generally begin from an already prepared analytical dataset. They do not jointly evaluate the path from an auditable business event, through a reproducible historical snapshot, to the prediction and later action outcome. For the target SME, this missing path matters because a model cannot be reproduced if invoice corrections, stock movements, organization scope and feature cutoffs are not traceable.

G3—Methodological-validity gap. Prior work separately establishes the risks of leakage [21], intermittent demand [9, 34], incoherent hierarchical forecasts [18], unsuitable error measures [19] and uncalibrated classification probabilities [29]. What is missing is a single SME evaluation protocol that applies these safeguards together: decision-time feature availability, temporal/entity-safe splits, baseline eligibility by series type, coherent aggregation, calibration and an untouched final test.

G4—Prediction-to-decision gap. Accurate forecasts or customer rankings are intermediate outputs. Stock-control and profit-aware churn studies demonstrate that operational consequences can change model choice [33, 35]; prescriptive analytics formalizes the use of predictions inside a decision problem [1]. The unresolved need is a transparent, low-resource mechanism that converts the outputs of several SME analytical tasks into actions while checking cash, stock, lead time, order rules, contact consent, uncertainty and action cost.

G5—Owner-centred explanation gap. SHAP provides principled feature attributions for complex predictions [23, 24], while human-explanation research shows that useful explanations are selective and contextual [27]. Neither technical attribution alone nor a generic dashboard establishes that a Bangladeshi SME owner understands the evidence, distinguishes association from cause, and completes the intended decision correctly. Explanation quality must therefore be evaluated as owner task performance, not inferred from the presence of an explanation chart.

G6—Lifecycle-accountability gap. Concept-drift literature explains how relationships between inputs and outcomes may change [12], and design science requires evaluation of the artifact in relation to its problem [15]. The reviewed task studies do not collectively close the local loop by recording model/baseline version, data cutoff, recommendation, owner response, later outcome and revalidation decision. Without that record, a system cannot show whether a recommendation remained appropriate after business conditions changed.

Central research gap. Within the reviewed evidence, there is no jointly evaluated, Bangladesh-oriented retail SME framework that connects traceable operational records to leakage-controlled and calibrated multi-task analytics, then converts those outputs into constraint-aware, owner-understandable actions whose use and outcomes can be monitored over time.

The central gap is integrative and evaluative. It does not require inventing a new classifier; it requires showing that established methods can be combined without breaking data lineage, temporal validity, decision feasibility or human accountability. Table 2.3 makes each gap falsifiable by linking it to a proposed response and evaluation.

TABLE 2.3: Traceability from identified gap to proposed response and test.

Gap	Proposed response	Evidence required to close the gap	Study link
G1: Context	Tier-1 Bangladesh SME study with explicit external-data boundaries	Primary local records and firm descriptions; no transfer of foreign/generated claims	RQ1–RQ4
G2: Integration	Canonical events, manifests and versioned historical snapshots	Reproducible trace from source row to feature, output, action and later outcome	O1, O2
G3: Validity	Temporal/entity-safe splits, baseline ladder, calibration and frozen test	Out-of-time metrics, ablations, intervals and leakage audit	RQ1, RQ2
G4: Decision	Constraint and expected-utility layer for reorder/retention actions	Same-path policy comparison and sensitivity to cost assumptions	RQ3
G5: Explanation	Selective Bangla evidence with source and uncertainty	Counterbalanced owner tasks measuring correctness, time and confidence	RQ4
G6: Life-cycle	Version, drift, response and outcome monitoring protocol	Revalidation log and post-decision observations	O6

B-SMART is positioned as the proposed response to this combined gap. Public or generated evidence can test parts of the procedure; closing the contextual, decision and owner-centred gaps requires consenting SME data and participant evaluation. If those data are unavailable, the final thesis must report the corresponding gaps as still open rather than treating a prototype or external benchmark as proof of local effectiveness.

Chapter 3

Proposed Methodology

3.1 Research Approach

The thesis will use design science with empirical evaluation. Design science is appropriate because the research produces an artifact intended to address a practical organizational problem and evaluates that artifact against explicit requirements [15]. The artifact is the B-SMART framework, including its canonical data design, time-aware analytical pipeline, constraint-aware recommendation procedure and owner-facing evidence. Empirical study is required because a plausible architecture alone cannot establish predictive validity, usability or business value.

The study is organized into four connected phases:

- P1. **Problem and evidence definition:** establish the target SME setting, research questions, literature-derived requirements, permitted claims and success measures.
- P2. **Artifact design:** specify the operational schema, historical snapshots, candidate analytical methods, strategy constraints, interfaces and audit records.
- P3. **Technical evaluation:** test data integrity, prediction, calibration, robustness and scalability using strictly separated development, public-real and primary-real evidence.
- P4. **Decision and user evaluation:** assess simulated or pilot inventory consequences and test whether owners can understand and correctly use the recommendations.

The unit of forecasting is a branch-SKU-day or branch-SKU-week, selected according to data density. The unit of customer-risk modelling is a customer's feature snapshot at a historical cutoff. The unit of strategy evaluation is an eligible action evaluated under the same information and constraints as its comparison policy.

3.2 Research Questions, Hypotheses and Measures

The research questions introduced in Chapter 1 are operationalized as follows:

RQ1 / H1.

A demand model containing local calendar, price, promotion and available market-context features will reduce out-of-time WAPE or MASE relative to the otherwise identical model without those feature groups and relative to a seasonal-naive baseline.

RQ2 / H2.

A model trained from historical customer snapshots will identify future inactivity better than the event base rate, measured by PR-AUC, and will achieve Lift@10% above 1.0 while retaining acceptable probability calibration.

RQ3 / H3.

Under identical demand sequences, costs and constraints, the proposed forecast-driven replenishment policy will reduce total stock-out and holding cost relative to the participating owner’s documented current policy.

RQ4 / H4.

In counterbalanced decision tasks, an explanation condition will improve task correctness or confidence without a practically harmful increase in completion time compared with an output-only condition.

TABLE 3.1: Alignment of research questions, evidence and primary measures.

RQ	Unit and comparison	Primary measures	Required evidence
RQ1	Unseen branch-SKU periods; context model versus ablations/baselines	WAPE, MASE, interval coverage	Time-stamped sales, stock, price, promotion and calendar data
RQ2	Future customer windows; model versus base-rate/rule baseline	PR-AUC, Lift@10%, Brier score, calibration	Pseudonymous customer transactions with historical cutoffs
RQ3	Same demand path; proposed versus current replenishment policy	Stock-out units, holding units, total cost	Stock, lead time, order constraints and validated cost assumptions
RQ4	Same decision scenarios; explanation versus output only	Correctness, time, confidence, perceived usability	Consenting target users and counterbalanced tasks

Hypotheses and primary measures are fixed before the final test is inspected. Secondary analyses will be labelled as exploratory. A statistically detectable difference will not be described as practically important unless its magnitude or cost implication is material.

3.3 Target Population and Sampling Plan

The target population comprises inventory-based retail SMEs operating in Bangladesh. To reduce uncontrolled variation, initial recruitment will focus on three to five businesses from one broadly comparable retail vertical. The planned analytical target is 12–24 months of records across the partners, subject to ethics approval and data availability. A first partner with a shorter history may be used only to test collection, mapping and data-quality rules.

Businesses will be recruited purposively because access to longitudinal invoice and stock data requires owner agreement. Inclusion requires identifiable dates, products and quantities and a person able to explain the current replenishment process. Customer-level analysis additionally requires stable pseudonymous customer identifiers and sufficient follow-up after each cutoff. A business or task will be excluded from a particular analysis when these minimum conditions are absent; it will not be repaired by inventing identifiers or outcomes.

3.4 Data Sources and Evidence Tiers

Evidence will be kept in separate tiers so that convenience data do not silently change the claim being tested.

TABLE 3.2: Planned evidence tiers and claim boundaries.

Tier	Source	Permitted use	Prohibited inference
1	Consenting Bangladesh SME invoice, stock and operating records	Primary local forecasting, customer, strategy and usability evaluation	Generalization beyond sampled firms without further evidence
2a	Bangladesh WFP/DAM market-price data [36]	Local external context and separate price-series study	SKU sales, customer behavior or firm revenue
2b	UCI Online Retail II, United Kingdom [5]	Real-invoice pipeline and method portability checks	Bangladesh purchasing behavior
2c	Public Bangladesh retailer demand series [20]	Local quantity-forecast procedure check	Multi-SKU, customer, price or profit claims if fields are absent
3	Clearly labelled generated records	Software tests, rare cases, scale tests and privacy-safe examples	Field effectiveness or real customer behavior

Every received dataset will have a manifest containing source, owner or repository, collection period, receipt date, checksum, license or agreement, row count, schema version,

mapping, exclusions and known limitations. Tier-3 data will never be merged into a Tier-1 test set. When augmentation is examined, the real-only and augmented training conditions will be reported separately and evaluated on the same untouched real test period.

3.5 Data Acquisition and Variable Plan

Primary collection will prefer exports from the business’s existing point of sale or spreadsheet workflow. A controlled template will be offered when no consistent export exists. The minimum sales fields are timestamp, business, branch, invoice, SKU, quantity, unit price, discount and cancellation/return status. Recommended fields include unit cost, promotion, payment mode, pseudonymous customer, stock-on-hand, stock movement, purchase receipt, supplier lead time and incoming quantity.

Variables are organized by the decision time at which they become available:

TABLE 3.3: Feature availability and leakage rule.

Availability class	Examples	Rule
Known before cutoff	Historical sales, current stock, recorded price, lead time, calendar	Eligible if timestamp and provenance are valid
Known for forecast horizon	Confirmed promotion, planned closure, calendar event	Eligible only when the same information would exist in real use
Observed after outcome	Later purchase, delivered satisfaction, realized return, future stock adjustment	Label or evaluation outcome only; never a predictor
Uncertain or revised	Unconfirmed promotion, estimated cost, manual lead time	Retain source and uncertainty; include a sensitivity analysis

3.6 Ethics, Privacy and Governance

University ethics approval and a signed data-sharing agreement will precede primary collection. Direct identifiers such as names, phone numbers, e-mail addresses, national identifiers, account numbers and wallet numbers are not required for the research tables. Each business will use a separate salted pseudonymous customer key; any re-identification mapping will remain with the business and outside the research repository.

The agreement will state research purpose, authorized researchers, storage, retention deadline, publication form, withdrawal and deletion procedure. Access will follow least privilege, and published material will use aggregates or separately reviewed de-identified examples. A retention suggestion will not authorize contact: customer consent and the

participating business's policy are independent feasibility constraints. Model outputs will support, not replace, owner decisions.

3.7 Canonical Operational and Analytical Model

The proposed operational schema contains Organization, Branch, Product, Customer, Supplier, SalesOrder, SalesOrderItem, Payment, InventoryBalance, StockMovement, PurchaseOrder, Expense, LedgerEntry, SalesReturn, Refund, ImportBatch and AuditLog entities. The analytical layer does not train directly on mutable operational tables. It creates versioned invoice-line, SKU-time and customer-cutoff views with the source keys and data cutoff retained.

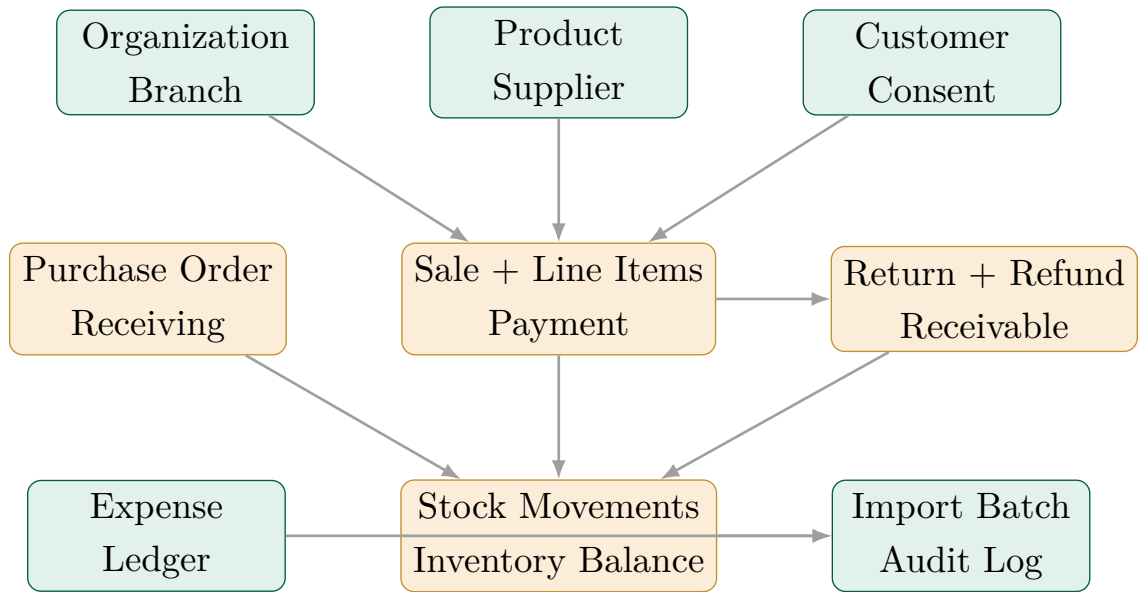


FIGURE 3.1: Proposed canonical operational data model.

Validation will test uniqueness of business-scoped identifiers, chronological consistency, nonnegative physical quantities where appropriate, payment/order reconciliation, cancellation handling and stock-movement balance. Corrections will be recorded as transformations rather than silently overwriting the source.

3.8 Formal Problem Formulation

For business b , branch l , product p , customer c and time t , let

$$D_t = \{\text{Sales, Stock, Purchases, Expenses, Customers, Payments, Context}\}_{\leq t}. \quad (3.1)$$

3.8.1 Performance Evaluation

For each KPI K , the system returns $K(D_t; w, g)$, where w is the reporting window and g is the aggregation level. Each value retains numerator, denominator, filters and data-quality status so that the owner can reproduce it.

3.8.2 Demand Prediction

$$\hat{y}_{b,l,p,t+h} = f\left(X_t^{\text{history}}, X_t^{\text{price}}, X_t^{\text{promotion}}, X_{t:t+h}^{\text{calendar}}, X_t^{\text{market}}\right), \quad (3.2)$$

where $\hat{y}$ is expected demand at horizon h . Quantile forecasts or calibrated residual intervals will represent uncertainty. When product-level forecasts are aggregated, reconciliation will be considered so SKU, category, branch and business totals remain coherent [18].

3.8.3 Future Customer Inactivity

$$q_{c,t} = P(\text{no purchase in } (t, t + H] \mid X_{c,t}), \quad (3.3)$$

where $X_{c,t}$ contains only records available by cutoff t . Horizon H will be fixed from business purchase cycles before final evaluation.

3.8.4 Strategy Optimization

For an eligible action $a \in A_t$,

$$U(a \mid D_t) = E[\Delta\text{Profit}(a)] - \text{ActionCost}(a) - \lambda \text{Risk}(a), \quad (3.4)$$

subject to budget, storage, minimum order quantity, supplier lead time, current/incoming stock and contact consent. This is a transparent operational objective inspired by the predictive-to-prescriptive transition [1]; its terms will be estimated and sensitivity tested rather than presented as known constants.

3.9 Preprocessing and Historical Snapshots

The processing order is fixed: validate source and schema; remove or mark exact duplicates; resolve cancellations and returns; construct the split; fit imputation, encoding and scaling on training data; build features; train; and evaluate once on the untouched test. The order prevents statistics from future periods entering training [21].

Demand features may include lags, rolling location and variability, day of week, month, festival distance, price, discount, known promotion, stock-out indicator and external market context. Zero demand caused by no sale is distinguished, when possible, from zero observed sales caused by no stock. Intermittency will be measured before choosing temporal granularity and candidate methods.

Customer snapshots will include RFM, tenure, order-value statistics, category breadth, purchase interval, discount behavior and pre-cutoff return rate. The label window follows the observation window, with a purge interval when feature construction or delayed events could overlap it. No customer identifier appears on both sides of an entity-generalization test.

3.10 Candidate Methods and Selection Rules

Method selection follows a baseline ladder, and eligibility depends on evidence volume and task structure.

- **Performance:** auditable KPI definitions and rule-based anomaly flags establish the descriptive baseline.
- **Demand:** last-value, seasonal-naive, moving-average or exponential-smoothing baselines precede ARIMA/SARIMA, Croston-family methods, gradient-boosted lag models and, only with adequate sequences, LSTM. Croston and the Syntetos–Boylan correction address intermittent demand [9, 34].
- **Inactivity:** a recency or base-rate rule is compared with Logistic Regression, Random Forest [2] and XGBoost [7]. Class weighting and decision threshold are selected on development data.
- **Segmentation:** scaled RFM features are studied with K-Means. Candidate K values are judged by inertia, silhouette, resampling stability and business interpretability.
- **Explanation:** coefficient/rule explanations are preferred for simple models; SHAP supplies model-behavior attributions for tree models [23, 24]. Displayed reasons will remain selective and contextual [27].

A complex candidate will be rejected when it lacks sufficient training history, cannot beat the relevant baseline on development data, produces unstable probabilities, or adds operational cost without material value. This rule keeps the proposed system useful when the scientifically honest answer is a simple forecast or business rule.

3.11 The B-SMART Sequential Procedure

Algorithm 1: B-SMART Strategy Recommendation

Input: Data available at cutoff D_t , horizon H , constraints C_t , validated model/baseline versions M , candidate actions A

Output: Ranked feasible recommendations R_t

- 1 Verify source, schema, timestamps and organization ownership;
 - 2 Create the historical feature snapshot X_t ;
 - 3 Apply transformations fitted on the appropriate training evidence;
 - 4 Estimate demand with uncertainty and customer inactivity probabilities;
 - 5 Calculate reproducible KPIs and data-quality warnings;
 - 6 Generate candidate actions with source and decision-time metadata;
 - 7 **foreach** $a \in A$ **do**
 - 8 Check feasibility under C_t ;
 - 9 Estimate benefit, cost and uncertainty using declared assumptions;
 - 10 $U(a) \leftarrow E[\Delta\text{Profit}(a)] - \text{Cost}(a) - \lambda\text{Risk}(a)$;
 - 11 Remove infeasible actions and rank the remainder by $U(a)$;
 - 12 Attach concise explanation, uncertainty, cutoff and evidence source;
 - 13 Return top- K actions; record owner response and later observable outcome;
-

For N invoice lines, d validated fields and P candidate actions, source validation and feature aggregation are approximately $O(Nd)$ and action ranking is $O(P \log P)$. Training complexity depends on the selected model and is measured empirically. Serving reuses versioned artifacts; it must not retrain in the request path.

3.12 Proposed Architecture and Information Flow

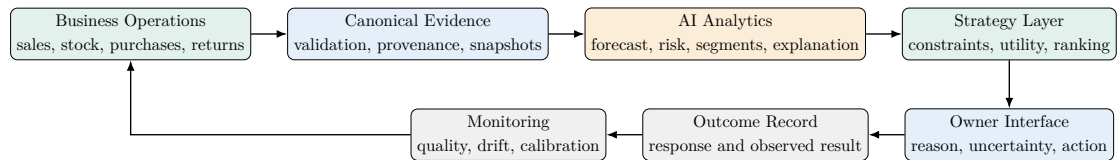


FIGURE 3.2: Proposed B-SMART architecture and accountable feedback loop.

The information flow is intentionally one-directional at prediction time: future outcomes cannot alter the snapshot that produced an earlier result. Outcomes enter later through the monitoring path and become eligible training evidence only in a new, versioned cycle. This separation makes it possible to reproduce what the system knew when an action was proposed.

3.13 Evaluation Protocol

3.13.1 Forecasting Evaluation

All forecasting splits preserve time. Candidate methods are selected through rolling-origin or expanding-window validation, followed by one evaluation on a final untouched period. For actual values y_i and forecasts $\hat{y}_i$,

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (3.5)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (3.6)$$

$$\text{WAPE} = \frac{\sum_i |y_i - \hat{y}_i|}{\sum_i |y_i|}. \quad (3.7)$$

MASE will support comparison across scales and avoid some defects of percentage errors [19]. Prediction-interval coverage and width or a proper scoring rule will evaluate uncertainty [13]. Results will be stratified by demand volume and intermittency rather than hiding failure on low-volume items in one aggregate.

3.13.2 Customer and Segmentation Evaluation

Inactivity models will report outcome prevalence, PR-AUC, ROC-AUC, Lift@10%, Brier score and reliability plots. The emphasis on PR measures follows their value under imbalance [31]; calibration is assessed separately because a useful ranking can still produce misleading probabilities [29]. Thresholds and contact fractions are chosen on validation evidence using declared costs, not adjusted after viewing the test.

Clustering evaluation will report inertia, silhouette, cluster stability, minimum cluster size and RFM profiles. The final interpretation will be reviewed with domain participants. A low-separation result will be reported as such; the existence of clusters will not be treated as proof of natural customer types.

3.13.3 Ablation, Robustness and Statistical Analysis

Planned ablations remove one feature family at a time: local calendar, price/promotion, market context and customer-history extensions. Additional comparisons examine global versus business-specific models and real-only versus clearly controlled augmentation. All paired comparisons use the same test instances or forecast origins.

Uncertainty will be summarized with paired block bootstrap intervals across time or business units, as appropriate. Model differences will be reported with effect size and interval, not only a significance label. Sensitivity analysis will vary lead time, shortage cost, holding cost, action cost and risk penalty in Equation 3.4.

3.13.4 Decision, Usability and Scalability Evaluation

Replenishment policies will face identical observed or replayed demand paths. The comparison records stock-out units, average inventory, holding cost, shortage cost and total cost. Retention ranking will use expected value only when response benefit, contact cost and consent are available; otherwise it will remain a prioritization study, not a profit claim [35].

Owner evaluation will use realistic tasks in a counterbalanced within-subject design when the sample permits. Measures include task correctness, completion time, error count, confidence and perceived usability. Participants may explain their decision in their own words, allowing the study to detect plausible but misunderstood explanations.

Scalability tests will use generated data only and will remain separate from accuracy results. Planned sizes are 100, 500, 1,000, 10,000, 100,000 and 1,000,000 transaction lines. Measures include import and preprocessing time, training time, prediction latency, API throughput, peak memory, artifact size and representative query latency. Any association-rule support experiment will be reported only after the corresponding algorithm exists and has been run.

3.13.5 Monitoring Plan

Monitoring will distinguish schema or data-quality failure, input shift, calibration change and outcome-performance change. Alerts will not automatically authorize retraining. A new model must repeat the validation protocol and retain its training window, features, parameters and comparison result. This follows the concept-drift literature's emphasis on both adaptation and appropriate evaluation [12].

3.14 Threats to Validity and Planned Controls

TABLE 3.4: Principal validity threats and planned controls.

Threat	How it may distort evidence	Planned control
Selection bias	Cooperative firms may be more digitally mature	Describe recruitment and firms; restrict generalization
Leakage	Future or target-derived data inflate performance	Availability table, cutoff snapshots, grouped/temporal split
Stock-out censoring	Observed sales understate unconstrained demand	Retain stock state; stratify or flag censored periods
Small/short series	Complex models overfit	Eligibility rule, simple baselines, rolling validation
Dataset mismatch	Foreign/public tasks are mistaken for local evidence	Evidence tiers and separate result tables
Uncertain costs	Utility ranking depends on weak assumptions	Owner-validated ranges and sensitivity analysis
Novelty/learning effect	Interface condition benefits from presentation order	Counterbalance tasks and collect task-level timing
Concept drift	Festival, price or behavior relations change	Time-based tests and post-deployment monitoring plan

3.15 Two-Semester Work Plan

The project period is February 2026–February 2027. The first semester defines the problem, evidence base and proposed artifact; the second emphasizes primary data, implementation completion, controlled evaluation and consolidation. The schedule in Table 3.5 is dependency-aware: the final model study cannot precede ethics/data readiness, and the final conclusions cannot precede the frozen evaluation.

TABLE 3.5: Project timeline from February 2026 to February 2027.

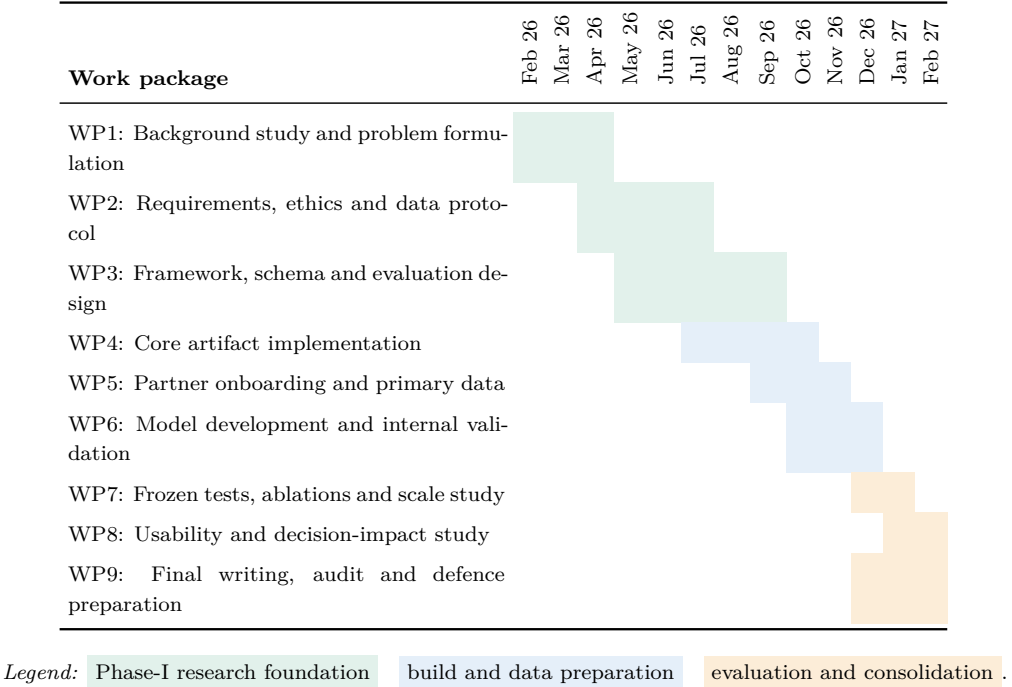


TABLE 3.6: Milestones, exit criteria and contingency.

Milestone	Target	Exit criterion	Contingency
M1: Initial re-port	September 2026	Problem, verified background, proposed method and approved plan	Revise scope/RQs before model lock
M2: Data readiness	November 2026	Ethics/agreement complete; manifest and quality report for eligible primary data	Narrow the primary task; retain public data only as external procedure checks
M3: Model freeze	December 2026	Baselines, features, splits, metrics and seeds fixed before final test	Prefer simpler eligible models when volume is inadequate
M4: Evaluation freeze	January 2027	Untouched tests, ablations, intervals, decision and scale analyses complete	Report inconclusive/negative findings; do not retune on test
M5: Final thesis	February 2027	Chapters complete; citations, claims, figures and reproducibility artifacts audited	Remove unsupported claims and document remaining limitations

The critical path runs through ethics and primary-data readiness (M2), model freeze (M3) and evaluation freeze (M4). Weekly progress review will track a deliverable, its evidence and its next dependency. Schedule pressure will not be handled by converting a public or generated-data result into a primary local claim.

3.16 Expected Methodological Outputs

At completion, the methodology is expected to yield a documented canonical schema and data dictionary, partner data protocol, versioned feature snapshots, baseline and candidate model specifications, a constraint-aware recommendation procedure, a reproducible evaluation package and an owner-study instrument. Implementation detail, observed numerical results and conclusions will be added to the reserved chapters only after the corresponding milestones are satisfied.

Chapter 4

Implementation

Chapter 5

Experimental Results

Chapter 6

Conclusions

Bibliography

- [1] BERTSIMAS, D., AND KALLUS, N. From predictive to prescriptive analytics. *Management Science* 66, 3 (2020), 1025–1044.
- [2] BREIMAN, L. Random forests. *Machine Learning* 45 (2001), 5–32.
- [3] BUREZ, J., AND VAN DEN POEL, D. Handling class imbalance in customer churn prediction. *Expert Systems with Applications* 36, 3 (2009), 4626–4636.
- [4] CARBONNEAU, R., LAFRAMBOISE, K., AND VAHIDOV, R. Application of machine learning techniques for supply chain demand forecasting. *European Journal of Operational Research* 184, 3 (2008), 1140–1154.
- [5] CHEN, D. Online retail ii. UCI Machine Learning Repository, 2012. CC BY 4.0; accessed 6 September 2026.
- [6] CHEN, H., CHIANG, R. H. L., AND STOREY, V. C. Business intelligence and analytics: From big data to big impact. *MIS Quarterly* 36, 4 (2012), 1165–1188.
- [7] CHEN, T., AND GUESTRIN, C. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (2016), pp. 785–794.
- [8] CHENG, C.-H., AND CHEN, Y.-S. Classifying the segmentation of customer value via RFM model and RS theory. *Expert Systems with Applications* 36, 3 (2009), 4176–4184.
- [9] CROSTON, J. D. Forecasting and stock control for intermittent demands. *Operational Research Quarterly* 23, 3 (1972), 289–303.
- [10] FADER, P. S., HARDIE, B. G. S., AND LEE, K. L. RFM and CLV: Using iso-value curves for customer base analysis. *Journal of Marketing Research* 42, 4 (2005), 415–430.
- [11] FILDES, R., MA, S., AND KOLASSA, S. Retail forecasting: Research and practice. *International Journal of Forecasting* 38, 4 (2022), 1283–1318.
- [12] GAMA, J., ŽLIOBAITĖ, I., BIFET, A., PECHENIZKIY, M., AND BOUCHACHIA, A. A survey on concept drift adaptation. *ACM Computing Surveys* 46, 4 (2014), 44:1–44:37.

- [13] GNEITING, T., AND RAFTERY, A. E. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 102, 477 (2007), 359–378.
- [14] HAZRA, U., AND PRIYO, A. K. K. Mobile financial services in bangladesh: Understanding the affordances. *The Electronic Journal of Information Systems in Developing Countries* 87, 3 (2021), e12166.
- [15] HEVNER, A. R., MARCH, S. T., PARK, J., AND RAM, S. Design science in information systems research. *MIS Quarterly* 28, 1 (2004), 75–105.
- [16] HOCHREITER, S., AND SCHMIDHUBER, J. Long short-term memory. *Neural Computation* 9, 8 (1997), 1735–1780.
- [17] HOQUE, M. R., SAIF, A. N. M., ALBAR, A. M., AND BAO, Y. Adoption of information and communication technology for development: A case study of small and medium enterprises in bangladesh. *Information Development* 32, 4 (2016), 986–1000.
- [18] HYNDMAN, R. J., AHMED, R. A., ATHANASOPOULOS, G., AND SHANG, H. L. Optimal combination forecasts for hierarchical time series. *Computational Statistics & Data Analysis* 55, 9 (2011), 2579–2589.
- [19] HYNDMAN, R. J., AND KOEHLER, A. B. Another look at measures of forecast accuracy. *International Journal of Forecasting* 22, 4 (2006), 679–688.
- [20] JAHIN, M. A., AND SHAHRIAR, A. Supply chain demand forecasting dataset of bangladeshi retailer. Mendeley Data, Version 1, 2024. CC BY 4.0; accessed 6 September 2026.
- [21] KAUFMAN, S., ROSSET, S., PERLICH, C., AND STITELMAN, O. Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data* 6, 4 (2012), 15:1–15:21.
- [22] KIM, S.-Y., JUNG, T.-S., SUH, E.-H., AND HWANG, H.-S. Customer segmentation and strategy development based on customer lifetime value: A case study. *Expert Systems with Applications* 31, 1 (2006), 101–107.
- [23] LUNDBERG, S. M., ERION, G., CHEN, H., DEGRAVE, A., PRUTKIN, J. M., NAIR, B., KATZ, R., HIMMELFARB, J., BANSAL, N., AND LEE, S.-I. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence* 2 (2020), 56–67.
- [24] LUNDBERG, S. M., AND LEE, S.-I. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems* 30 (2017), pp. 4765–4774.
- [25] MAKRIDAKIS, S., SPILIOTIS, E., AND ASSIMAKOPOULOS, V. The M4 competition: Results, findings, conclusion and way forward. *International Journal of Forecasting* 34, 4 (2018), 802–808.

- [26] MAKRIDAKIS, S., SPILIOTIS, E., AND ASSIMAKOPOULOS, V. The M5 competition: Background, organization, and implementation. *International Journal of Forecasting* 38, 4 (2022), 1325–1336.
- [27] MILLER, T. Explanation in artificial intelligence: Insights from the social sciences. *Artificial Intelligence* 267 (2019), 1–38.
- [28] NGAI, E. W. T., XIU, L., AND CHAU, D. C. K. Application of data mining techniques in customer relationship management: A literature review and classification. *Expert Systems with Applications* 36, 2 (2009), 2592–2602.
- [29] NICULESCU-MIZIL, A., AND CARUANA, R. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning* (2005), pp. 625–632.
- [30] ROUSSEEUW, P. J. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics* 20 (1987), 53–65.
- [31] SAITO, T., AND REHMSMEIER, M. The precision–recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE* 10, 3 (2015), e0118432.
- [32] SHAHADAT, M. M. H., NEKMAHMUD, M., EBRAHIMI, P., AND FEKETE-FARKAS, M. Digital technology adoption in SMEs: What technological, environmental and organizational factors influence SMEs’ ICT adoption in emerging countries? *Global Business Review* (2023), 1–27.
- [33] SYNTETOS, A. A., BABAI, M. Z., DAVIES, J., AND STEPHENSON, D. Forecasting and stock control: A study in a wholesaling context. *International Journal of Production Economics* 127, 1 (2010), 103–111.
- [34] SYNTETOS, A. A., AND BOYLAN, J. E. The accuracy of intermittent demand estimates. *International Journal of Forecasting* 21, 2 (2005), 303–314.
- [35] VERBEKE, W., DEJAEGER, K., MARTENS, D., HUR, J., AND BAESSENS, B. New insights into churn prediction in the telecommunication sector: A profit driven data mining approach. *European Journal of Operational Research* 218, 1 (2012), 211–229.
- [36] WORLD FOOD PROGRAMME, AND DEPARTMENT OF AGRICULTURAL MARKETING, GOVERNMENT OF BANGLADESH. WFP food prices database – bangladesh. Humanitarian Data Exchange, 2026. CC BY-IGO; accessed 6 September 2026.